A 5 MS/s 10-Bit Asynchronous SAR ADC with a Two-Stage Dynamic Comparator in 65 nm CMOS

Thawin Serivivatanavongse
School of Electrical and Computer Engineering
Georgia Institute of Technology
Atlanta, USA
tseriviv3@gatech.edu

Passapan Sanguanchua
Department of Electrical and Computer Engineering
University of Illinois, Urbana-Champaign
Champaign, USA
ps96@illinois.edu

***Abstract*—This report presents the design and schematic-level characterization of a 10-bit, 5 MS/s successive approximation register analog-to-digital converter in a 65 nm CMOS process with a 1 V supply. The converter uses common-mode-based capacitor switching, a binary-weighted capacitive DAC, bootstrapped sampling switches, and a self-timed asynchronous conversion loop driven by a two-stage dynamic comparator. The comparator was characterized for speed, energy, hysteresis, input-referred noise, offset, and kickback across process and temperature corners, and several circuit modifications were introduced to suppress kickback and leakage. At the system level, the converter achieves 54.77 dB SNDR, 8.81 effective bits, and 62.99 dBc SFDR, with a maximum INL of 1.06 LSB and a maximum DNL of 0.49 LSB, while consuming 126.4 microwatts at the typical corner. A delta-length capacitor layout is also proposed to realize sub-femtofarad unit capacitors in future iterations.**

***Index Terms*—SAR ADC, dynamic comparator, asynchronous logic, kickback, common-mode-based switching, delta-length capacitor, 65 nm CMOS**

I. Introduction

Successive approximation register (SAR) ADCs have become one of the most energy-efficient and versatile data converter architectures. A SAR ADC performs a binary search: after sampling, each bit is resolved by comparing the held input against a DAC voltage that the SAR logic updates from the previous decisions. Because most of the circuit is digital or switched-capacitor and contains no static bias currents, SAR ADCs scale well with technology and occupy the medium-speed, medium-resolution space between sigma-delta and pipeline converters.

This report describes the design of a 10-bit, 5 MS/s SAR ADC in TSMC 65 nm CMOS with a 1 V supply, covering the flow from system-level architecture through transistor-level comparator design and schematic-level system characterization. The design follows the asynchronous SAR approach of Harpe *et al.* [2] and the area-optimized 10-bit design of Pelzers *et al.* [1], which serves as the main performance reference. The system uses V_{CM}-based switching [3] with a 1 fF unit capacitor. Table I summarizes the target specifications.

TABLE I
Design Specifications

Parameter	Target	Note
Resolution	10 bit	
Sample rate	5 MS/s	10 MS/s functional
Architecture	Async. SAR	
Technology	TSMC 65 nm	
Supply	1 V	
Unit capacitance	1 fF	125 aF in [1]
Input range	$\approx 1.8V_{DD}$	Set by C_u and C_p

II. System Overview

Fig. 1 shows the top-level architecture. A single-ended-to-differential converter produces $(V_{DD} \pm V_{IN})/2$, which is sampled onto the top plates of two capacitor DACs. The main blocks are:

- **Sampling switches** – bootstrapped switches that track and hold the differential input on the DAC top plates.
- **Capacitor DAC** – binary-weighted capacitor arrays whose bottom plates are switched to generate the binary-weighted test voltages at each bit decision.
- **Comparator** – a two-stage dynamic comparator that decides the sign of the top-plate differential voltage and generates its own conversion clock.
- **SAR logic** – stores each comparator decision, selects the next bottom-plate state, and outputs the final 10-bit code.
- **Bottom-plate drivers** – analog multiplexers that connect each capacitor bottom plate to V_{DD}, V_{CM}, or V_{SS}.
- **Equalizing switch** – shorts the two comparator inputs at the end of each conversion, removing residual charge and resetting the differential voltage to zero.

Each conversion proceeds in three phases. In the *sample* phase the input is tracked and held on the DAC top plates. In the *bit-cycling* phase the SAR logic tests one bit at a time from the MSB downward, halving the remaining search space with each comparison. In the *output* phase, after the LSB decision, the SAR logic holds the 10-bit code. With a 200 ns sample period, 150 ns is allotted to the conversion phase.

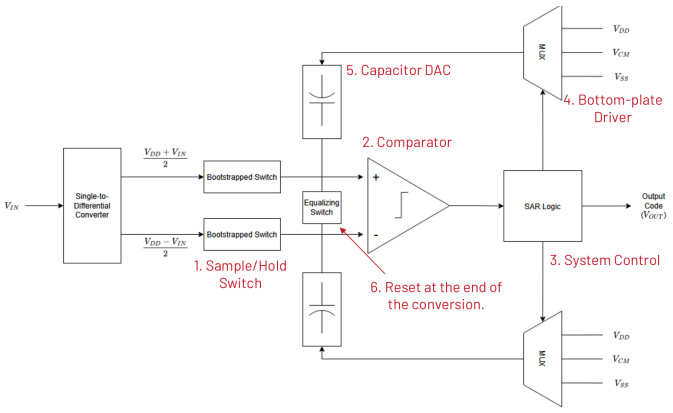


Fig. 1. Top-level architecture of the SAR ADC.

III. SAR Logic and V_{CM}-Based Switching

V_{CM}-based switching (also called merged capacitor switching) has been reported to offer lower switching energy and area than conventional and monotonic [4] switching schemes [3]. During standby and sampling, every bottom plate is held at the common-mode voltage $V_{CM} = V_{DD}/2 = 0.5$ V. Because the first comparison is made directly on the sampled input, no capacitor needs to be switched to resolve the MSB, so a 10-bit converter requires only nine binary-weighted capacitors ($256C_u$ down to C_u) plus a dummy C_u per side, for a total of $C_{tot} = 512C_u$.

After each decision, the bottom plate of the next-largest capacitor on one side is switched from V_{CM} to V_{DD} while the corresponding capacitor on the other side is switched to V_{SS}, moving the two top plates toward each other. The resulting top-plate step is

$$\Delta V_i = \frac{C_i}{C_{tot}} \cdot \frac{V_{DD}}{2}. \tag{1}$$

Because the two sides move in opposite directions by equal amounts, the input common-mode voltage of the comparator stays constant at V_{CM} throughout the conversion. This keeps the comparator operating point, offset, and noise independent of the code being resolved.

IV. Sampling and Bottom-Plate Switching Network

A. Bottom-Plate Driver

The bottom-plate driver connects each capacitor's bottom plate to one of three references. It consists of an NMOS pass transistor to V_{SS}, a PMOS pass transistor to V_{DD}, and a transmission gate to V_{CM}, since V_{CM} lies mid-rail and cannot be passed cleanly by a single device. A small decoder converts the two-bit select word from the SAR logic into the gate signals: code 00 selects V_{SS}, 01 selects V_{DD}, and 10 selects V_{CM}; code 11 is unused and leaves the output floating. The drivers were sized for 10 MS/s operation and could be further downsized for lower power, although their contribution is small relative to the other blocks.

B. Sampling Switch

The input is sampled through a bootstrapped switch, first introduced by Abo and Gray [5]. A charge-pump circuit holds the gate-to-source voltage of the sampling transistor at V_{DD} regardless of the input level, so the on-resistance

$$R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})} \tag{2}$$

becomes signal-independent [6]. This improves both linearity and tracking bandwidth compared with a simple pass gate. In this work the gate boosting is modeled with ideal multiplexed sources ($V_{IN}+V_{DD}$ during sampling, V_{SS} during hold), while the sampling transistor itself is a real device.

Two non-idealities affect the sampling switch: clock feedthrough and charge injection. Clock feedthrough was found to be negligible at the operating point in simulation. Charge injection is mitigated with a half-sized dummy transistor at the switch output that absorbs the channel charge released when the switch opens, at the cost of a marginal increase in parasitic capacitance.

C. Equalizing Switch

The equalizing switch is a modified bootstrapped switch connected between the two comparator inputs. At the end of each conversion it shorts the two top plates, flushing residual charge and resetting the differential voltage to zero so that no memory carries into the next sample. The switch is laid out symmetrically so that both halves of the differential DAC see identical loading.

V. Capacitor DAC

A. Unit Capacitance and Input Range

Ideal capacitors were used for the system-level simulations reported here. The choice of C_u balances input range against power and speed. Parasitic capacitance C_p at the top-plate node, dominated by the comparator input capacitance and switch junctions, attenuates the DAC steps and shrinks the input range:

$$V_{range} = \frac{C_{tot}}{C_{tot} + C_p} \cdot V_{FS}. \tag{3}$$

A smaller C_u or a larger C_p therefore reduces the range, and a larger C_u increases switching energy and the charge needed to settle the DAC. Conversely, (3) can be inverted to extract C_p from the measured range. With a 125 aF unit capacitor, as used in [1], the range in this design collapsed to roughly ± 0.6 V; with $C_u = 1$ fF it is ± 0.918 V, from which $C_p \approx 45.7$ fF.

B. Delta-Length Capacitor Layout

Sub-femtofarad capacitors cannot be drawn directly with acceptable accuracy in this node. Following [7], a delta-length layout is proposed in which each effective bit capacitance is the *difference* of two large metal-oxide-metal finger capacitors of lengths $L_0 + \Delta_i$ and $L_0 - \Delta_i$ that are switched in opposite directions:

$$C_{i,\text{eff}} = C_i - C_i'. \tag{4}$$

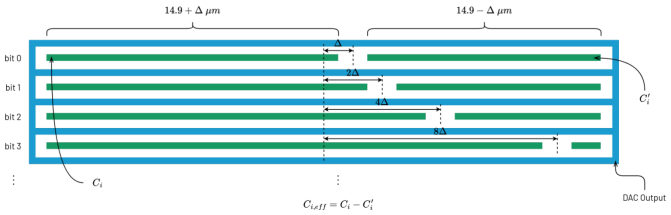


Fig. 2. Delta-length capacitor layout for the four LSBs (not drawn to scale). Each effective capacitance is the difference between a finger of length $14.9 + \Delta_i$ μm and one of $14.9 - \Delta_i$ μm.

TABLE II
EXTRACTED DELTA-LENGTH CAPACITANCES

Bit	Δ (μm)	C (fF)	C' (fF)	C_{eff} (fF)	Error (%)
0	0.2	4.589	4.462	0.127	–
1	0.4	4.585	4.353	0.232	8.66
2	0.8	4.699	4.223	0.476	6.30
3	1.6	5.012	4.066	0.946	6.89

Error relative to $2^i \cdot C_{\text{eff},0}$.

Binary weighting is achieved by scaling Δ_i, and edge and fringe effects common to both fingers cancel. The fingers are stacked on metals 6 and 7, so the coupling is strongly three-dimensional; the Quantus xACT field-solver engine, rather than the standard xRC extractor, is required for accurate extraction. Fig. 2 shows the layout concept and Table II lists the extracted values for the four LSBs.

The effective capacitance tracks the binary target within 6–9%.

VI. DYNAMIC COMPARATOR

A. Topology

The comparator is a two-stage (double-tail) dynamic comparator, following [8], chosen because it draws no static bias current. Fig. 3 shows the circuit. In the first stage, the differential input pair discharges the precharged nodes FN and FP at rates set by the input difference. In the second stage, FN and FP drive a cross-coupled latch through a PMOS input pair, and positive feedback regenerates the small first-stage imbalance into full-swing outputs SP and SN.

Two modifications were made to the original topology. First, the gate of the first-stage precharge PMOS pair is driven by a CHARGE signal, $\text{CHARGE} = \overline{\overline{\text{CLK}} \cdot \text{EN}}$, rather than the clock directly. CHARGE follows the conversion clock while the comparator is enabled and stays high when it is disabled, turning the precharge pair off between conversions. Second, an additional tail NMOS gated by $\overline{\text{EN}}$ holds the common-source node P at ground when the comparator is idle. Without it, subthreshold conduction through the input pair charges P during reset, and its rapid discharge at the onset of evaluation couples onto the input nodes. Fluctuation at node P is one of the main contributors to comparator kickback.

During sizing iterations, three further changes were made:

- The clocked tail device was changed to a high-threshold-voltage (HVT) transistor to reduce leakage while the comparator is idle. In simulation this also shortened the resolution time.
- The second-stage PMOS input pair was downsized to reduce the load on the cross-coupled latch, which sharpened regeneration of SP and SN.
- The width of the first-stage precharge PMOS pair was reduced to limit the slew rate of FN and FP, which reduced the peak kickback current from 102 μA to 66 μA (Section VI-H).

B. Asynchronous Clock Generation

The time a dynamic comparator needs to resolve grows as the input difference shrinks. A synchronous SAR must therefore allot every bit the worst-case period, wasting most of it on the early, large-input decisions. Instead, the comparator here generates its own clock through the self-oscillating loop of Fig. 4, so each bit takes only as long as it needs [2].

The ready signal $\text{RDY} = \overline{\overline{\text{SP}} \cdot \overline{\text{SN}}}$ goes high once either latch output resolves, and serves as the clock for the SAR logic. RDY and $\overline{\text{EN}}$ drive a NOR gate followed by two inverters to produce the comparator clock CLK, which resets the comparator and, after the loop delay, starts the next evaluation. The loop delay must leave enough time for the comparator to reset and for the DAC to settle before the next comparison. It was tuned by lengthening one of the inverters, which also delays RDY so that it aligns with the comparator outputs.

C. Speed Across Corners

Fig. 5 and Table III show the per-comparison clock period as a function of input difference. At a 1 LSB input, a comparison takes up to 3.2 times longer than at full scale. Summed over the 10 binary-weighted inputs of a conversion, the total comparison time is 44.3 ns at the slowest corner (SS, $-40\,^\circ$C), well within the 150 ns conversion budget. An empirical fit of the clock period versus input difference showed that a natural-logarithm model ($R^2 = 0.994$) describes the data far better than exponential or power-law models, consistent with the logarithmic dependence of latch regeneration time on initial imbalance. This model can be used in a behavioral comparator model.

The fast corner (FF, $100\,^\circ$C) was also checked for the opposite failure: a loop so fast that the comparator cannot fully reset between decisions. The comparator remained fully functional at this corner and at all other process and temperature combinations tested.

D. Energy per Comparison

Average power rises with input difference because larger inputs drive more current through the input pair, but large inputs also resolve faster. The energy per comparison is therefore the more meaningful metric. At the typical corner with 10 fF of added parasitic capacitance at FN, FP, SN, SP, and RDY, the energy is lowest, about 107 fJ, for medium inputs of 0.2–0.4 V and rises for very small inputs, where regeneration is slow. A sweep of the added parasitic capacitance from 1 to

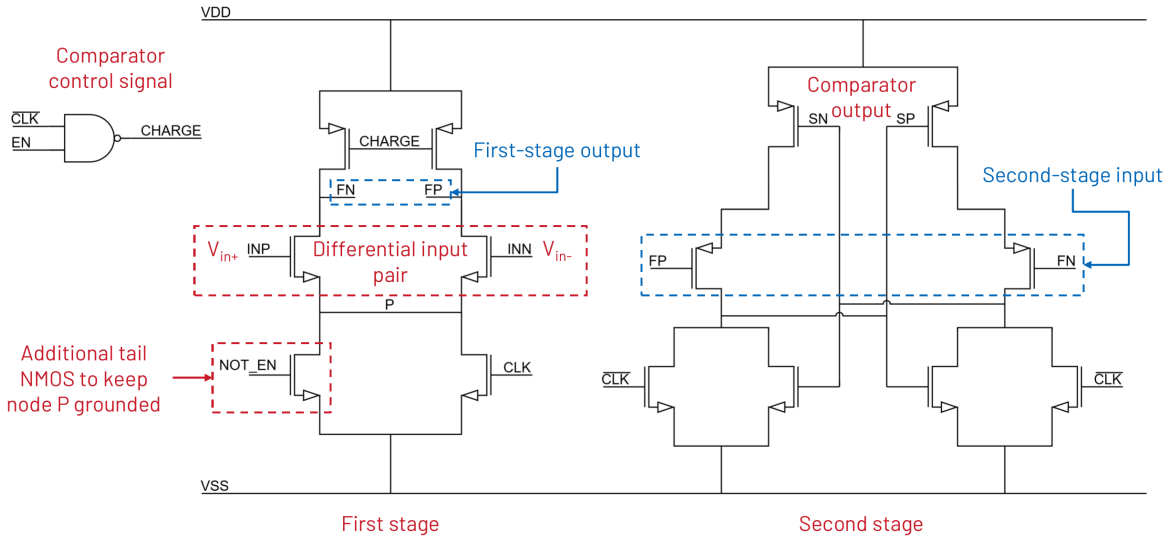


Fig. 3. Modified two-stage dynamic comparator. Left: first stage with CHARGE-controlled precharge and the additional $\overline{\text{EN}}$-gated tail device. Right: second-stage cross-coupled latch.

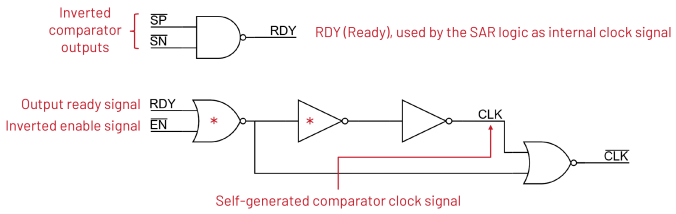


Fig. 4. Self-oscillating loop for asynchronous clock generation. Devices marked * use increased channel length for timing.

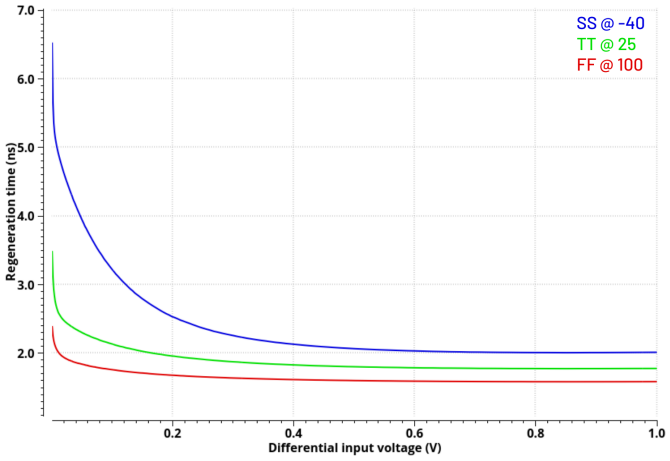


Fig. 5. Comparator regeneration time versus differential input at three corners.

20 fF showed that energy per comparison and clock period both increase approximately linearly with node capacitance, which highlights the importance of compact layout for this block.

TABLE III
AVERAGE COMPARATOR CLOCK PERIOD (NS)

V_{id} (mV)	SS, −40 °C	TT, 25 °C	FF, 100 °C
1 (≈1 LSB)	6.367	3.365	2.310
2	5.673	3.070	2.204
4	5.248	2.819	2.108
8	5.000	2.628	2.018
16	4.716	2.490	1.935
32	4.300	2.369	1.863
64	3.676	2.225	1.793
128	2.929	2.048	1.710
256	2.323	1.884	1.635
512	2.033	1.778	1.581
1024	1.987	1.758	1.567
Total, 10 bits	**44.252**	**26.434**	**20.724**

E. Hysteresis

Hysteresis is often added to stand-alone comparators to reject noise and prevent chatter, but in a SAR ADC it is detrimental. It makes the decision threshold depend on the previous output bit, which corrupts the final low-amplitude bit decisions and produces uneven code widths that increase DNL. Hysteresis was measured by slowly ramping the differential input from slightly negative to slightly positive and recording the trip voltage V_{TRIP}^{+}, then repeating with inverted inputs to obtain V_{TRIP}^{-}:

$$\begin{aligned} V_{HYS} &= V_{TRIP}^{+} - V_{TRIP}^{-} \\ &= 5.81\,\mu\text{V} - (-8.76\,\mu\text{V}) = 14.57\,\mu\text{V}. \end{aligned} \tag{5}$$

This is about 1.5% of an LSB, well below the common 10% rule of thumb [9], and is considered negligible.

F. Input-Referred Noise

Comparator noise was extracted from transient-noise simulations. At each input voltage, 200 Monte Carlo runs with

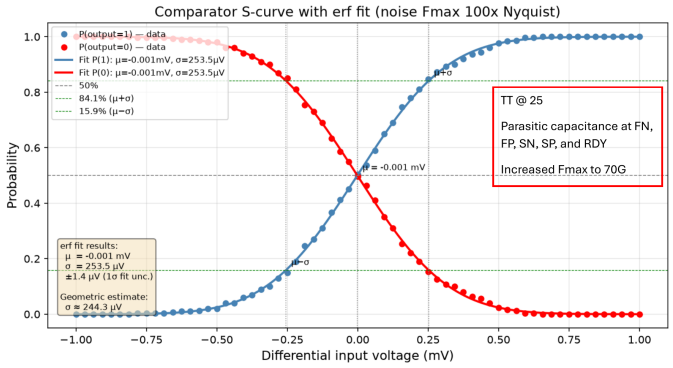


Fig. 6. Comparator S-curve with transient noise enabled (TT, 25 °C, 70 GHz noise bandwidth) and Gaussian fit, giving $\sigma_n = 253.5\,\mu$V.

TABLE IV
COMPARATOR INPUT-REFERRED NOISE

Corner	σ_n (μV)
SS, −40 °C	169.4
TT, 25 °C	253.5
FF, 100 °C	454.3
Quantization noise, $q/\sqrt{12}$	517

only transient noise enabled (channel thermal, flicker, gate-resistance, and source/drain-resistance noise) gave the probability of a "1" decision. Fitting the resulting S-curve with a Gaussian cumulative distribution function,

$$P(\text{out} = 1) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{V_{id} - \mu}{\sqrt{2}\,\sigma_n}\right)\right], \tag{6}$$

yields the input-referred noise σ_n directly, without requiring knowledge of the comparator's small-signal gain (Fig. 6).

The extracted σ_n depends on the noise bandwidth used in simulation. At the typical corner it increased from 155 μV with a 700 MHz noise bandwidth to 242 μV at 7 GHz and 254 μV at 70 GHz, indicating convergence near 250 μV. Table IV summarizes the corner results at the 70 GHz setting. The worst case, 454 μV at FF, 100 °C, is consistent with thermal noise dominating, since flicker noise and memory effects are suppressed by the fully dynamic operation [8]. For comparison, [8] reports 305 μV, and the quantization noise of this converter is $q/\sqrt{12} = 1.79\,\text{mV}/\sqrt{12} \approx 517\,\mu$V. At the typical corner, the comparator noise adds less than 1 dB to the total noise power.

G. Offset and Mismatch

A dynamic comparator works by racing two nominally identical branches against each other, so it is highly sensitive to device mismatch. Random dopant fluctuation and line-edge roughness cause uncorrelated device-to-device variation that widens the offset distribution, while global process variation shifts all devices together and mainly moves speed and power.

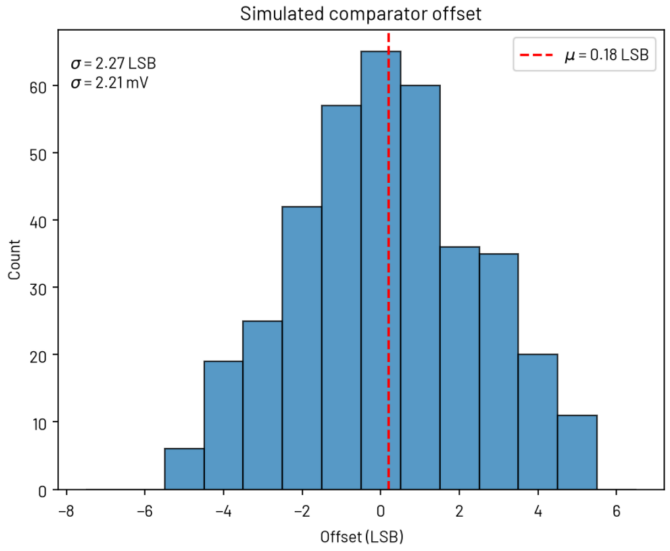


Fig. 7. Simulated comparator offset from 400 mismatch-only Monte Carlo runs.

TABLE V
COMPARATOR OFFSET COMPARISON

	[2]	This work
Resolution (bit)	8	10
Mean (LSB / mV)	−0.1 / −0.39	0.18 / 0.17
Std. dev. (LSB / mV)	0.8 / 3.13	2.27 / 2.21

The threshold-voltage mismatch of the input pair follows Pelgrom's law [10],

$$\sigma_{\Delta V_{TH}} = \frac{A_{VT}}{\sqrt{WL}}, \tag{7}$$

so offset can be reduced by enlarging the input pair. However, a larger pair increases the comparator input capacitance, which adds to C_p and shrinks the input range through (3), and it also increases kickback, power, and resolution time.

Offset was measured with mismatch-only Monte Carlo simulation. For each run, the differential input was swept and the trip point at which the output changes from 0 to 1 was recorded. Fig. 7 shows the distribution for the final comparator over 400 runs: $\mu_{os} = 0.17$ mV (0.18 LSB) and $\sigma_{os} = 2.21$ mV (2.27 LSB). An earlier sizing with a $32\,\mu\text{m}/1.6\,\mu\text{m}$ input pair achieved $\sigma_{os} = 0.89$ LSB over 500 runs, close to the 0.8 LSB of [2], but further enlargement gave diminishing returns relative to its cost in input capacitance. Table V compares the final result with [2]. In absolute terms, the offset spread is lower than that of the reference.

In a SAR ADC without redundancy, a static comparator offset shifts the whole transfer function by a fixed amount per die rather than introducing code-dependent errors. It therefore appears mainly as offset error, which can be calibrated, rather than as INL or DNL. A small auxiliary trim pair driven by static calibration voltages could null the offset without adding significant input capacitance.

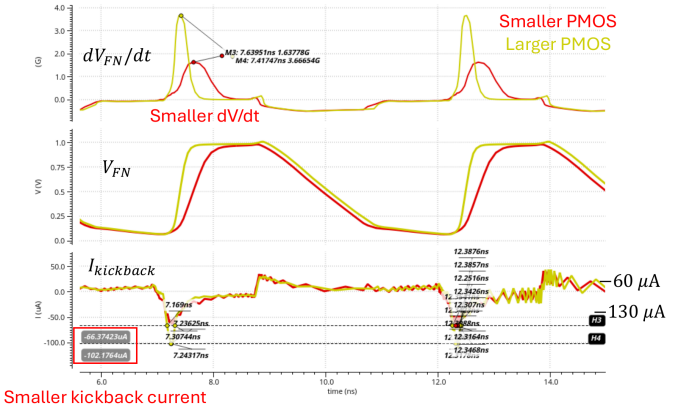


Fig. 8. Slew rate of FN, FN voltage, and kickback current for the original (yellow) and reduced-width (red) precharge PMOS pair.

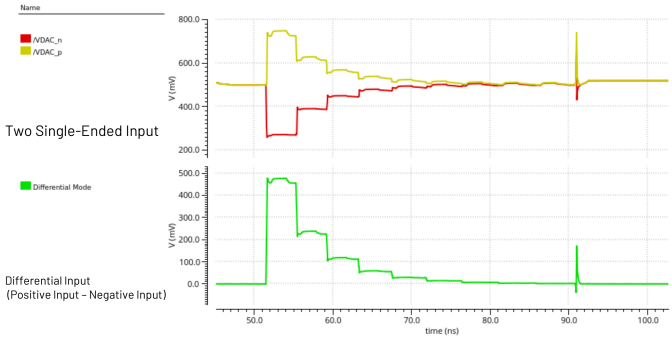


Fig. 9. DAC top-plate voltages (top) and differential voltage (bottom) during one conversion.

H. Kickback

When FN and FP swing during evaluation and reset, the signal is coupled back through the gate-drain capacitance of the input pair onto the floating DAC top plates. The kickback current is

$$I_{kb} = C_{gd}\,\frac{dV_{FN/FP}}{dt}, \tag{8}$$

which motivated reducing the precharge PMOS width to limit the slew rate. Fig. 8 shows the result: the slower FN transition lowers the peak kickback current from 102 μA to 66 μA.

However, the error on the DAC is set by the total injected charge,

$$Q_{kb} = \int C_{gd}\,\frac{dV_{FN/FP}}{dt}\,dt = C_{gd}\,\Delta V_{FN/FP}, \tag{9}$$

so reducing dV/dt lowers only the peak current, while reducing the voltage swing at the input-pair drain reduces the absolute kickback.

To reduce that swing, a cascode stage was investigated between the input pair and FN/FP. Biased at 0.4 V, below its threshold, the cascode operates in subthreshold and shields the input-pair drains (INTFN/INTFP) from the rail-to-rail swing at FN/FP. Equating the subthreshold cascode current to the input-pair current gives

$$V_{INTFN} = V_{bias} - V_{TH} + nV_T \ln\!\left(\frac{I_{0,CAS}}{I_{IN}}\right), \tag{10}$$

which self-regulates: if INTFN rises, the cascode can no longer supply the input-pair current and the node is pulled back down. In simulation, the drain swing fell from about 1 V to about 60 mV, roughly a 16-fold reduction in injected charge. A bottom cascode on the tail further suppressed the current surge that otherwise pulls the input-pair source and drains below ground at the start of evaluation.

The cascodes carry two costs. At a 1 mV input, the clock period rose from 5.35 ns to 7.55 ns. More seriously, at SS, $-40\,^{\circ}$C the cascode threshold rises by about 100 mV, the device cuts off, and the comparator becomes more than four times slower. A corner-tracking cascode bias would be required to make this technique robust, and the system results below use the comparator without cascodes.

At the system level (Fig. 9), kickback is signal-dependent and grows with the single-ended input level, but in the differential domain it is harmful only if it flips the sign of the differential input at the decision instant. For large differential inputs, the disturbance pushes the differential voltage further from zero at the decision point, which is benign. For small differential inputs, the ripple stays within 0.5 LSB of the settled value, which is acceptable.

VII. System-Level Results

The full ADC was simulated at the schematic level with transistor-level comparator, bottom-plate drivers, and sampling transistors, and with ideal SAR logic, gate-boosting sources, and capacitors. A 100 kHz sine input was first used to verify basic tracking. At the slowest corner (SS, $-40\,^{\circ}$C), the conversion finished with 70–80 ns of headroom, about 50% of the allotted conversion phase, which leaves room to raise the sample rate.

A. Static Metrics

Static performance was measured following IEEE Std 1241-2010 [11]. A slow linear ramp was applied and the output code transitions were extracted in Python post-processing; a slower ramp gives finer transition resolution at the cost of simulation time. The metrics are defined as follows:

- **Input range**: the difference between the first and last transition voltages.
- **Gain error**: the deviation of the best-fit line slope from ideal.
- **Offset error**: the shift of the central transition from ideal.
- **INL**: the deviation of each transition voltage from the best-fit line.
- **DNL**: the deviation of each code width from the measured LSB.

Fig. 10 shows the transfer function. The input range shrinks to ± 0.918 V, a gain error of -8.35%, which by (3) corresponds to about 45.7 fF of top-plate parasitic capacitance. The offset error is negligible at 0.023 LSB, the measured LSB is 1.79 mV, and the maximum INL and DNL are 1.06 LSB and 0.49 LSB, respectively (Table VI).

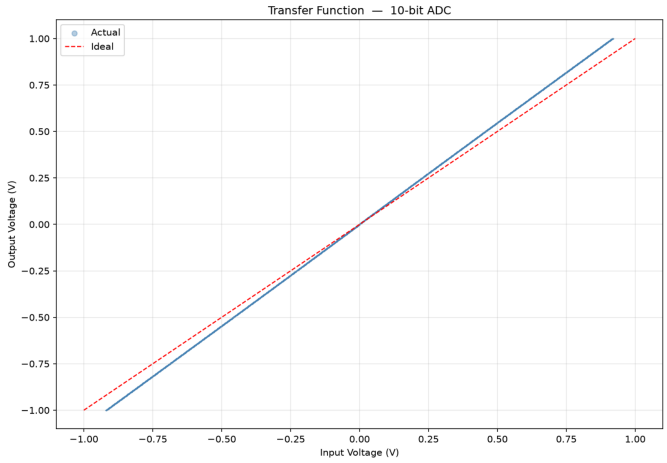


Fig. 10. Simulated ADC transfer function (blue) compared with the ideal (red dashed). The steeper slope reflects the reduced input range.

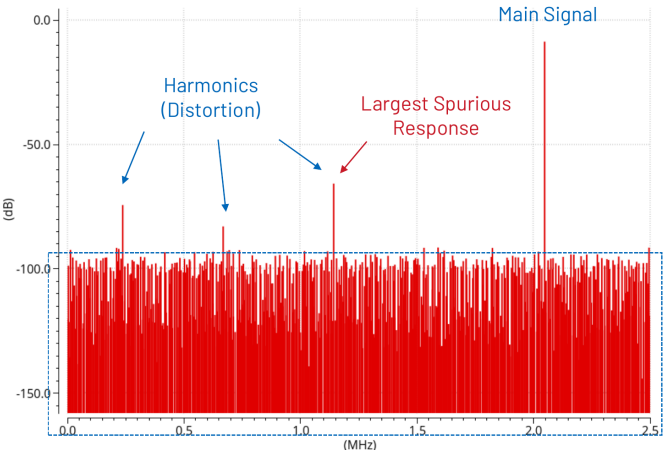


Fig. 11. Output spectrum for a pure-tone input, showing the harmonics and the largest spur that sets the SFDR.

B. Dynamic Metrics

Dynamic performance was evaluated from the FFT of the output for a pure-tone sine input (Fig. 11). The signal-to-noise-and-distortion ratio (SNDR) compares the signal power with all other spectral content, the SNR excludes harmonics, and the spurious-free dynamic range (SFDR) is the ratio of the signal to the largest spur. The effective number of bits follows from the SNDR:

$$\mathrm{ENOB} = \frac{\mathrm{SNDR}_{\mathrm{dB}} - 1.76}{6.02}. \tag{11}$$

The ADC achieves 54.77 dB SNDR (8.81 bits ENOB), 59.54 dB SNR, and 62.99 dBc SFDR, matching the SNDR and ENOB of [1], with SFDR about 2 dB lower.

C. Power and Figure of Merit

Table VII lists the total power at three corners and the Walden figure of merit,

$$\mathrm{FoM} = \frac{P}{f_s \cdot 2^{\mathrm{ENOB}}}, \tag{12}$$

TABLE VI
ADC Performance Summary

Parameter	This work	[1]
Static		
Input range	± 0.918 V	–
Top-plate parasitics	≈ 45.7 fF	–
Gain error	-8.35%	–
Offset error	0.023 LSB	–
Measured LSB	1.79 mV	–
Max INL	1.06 LSB	0.6 LSB
Max DNL	0.49 LSB	0.38 LSB
Dynamic		
SNDR	54.77 dB	54.7 dB
ENOB	8.81 bit	8.79 bit
SNR	59.54 dB	–
SFDR	62.99 dBc	65.1 dBc

"–" indicates a value not reported in [1].

TABLE VII
Power Consumption and FoM

Corner	Power (μW)	FoM (fJ/conv.-step)
SS, $-40\,^{\circ}$C	92.3	41.1
TT, $25\,^{\circ}$C	126.4	56.3
FF, $100\,^{\circ}$C	167.6	74.7

computed with $f_s = 5$ MS/s and the ENOB of 8.81 bits. At the typical corner the FoM is 56.3 fJ/conversion-step. Published SAR ADCs range from about 20 to 180 fJ/conversion-step [1], and state-of-the-art designs reach a few fJ/conversion-step [8], [12]. Two caveats apply: replacing the ideal components will add power and degrade performance, while the available speed headroom could be used to raise f_s and improve the FoM.

VIII. Future Work

The next steps toward a fabricated converter are:

- Replace the ideal blocks with transistor-level implementations: full bootstrapped switches, synthesized SAR logic, and the delta-length capacitor array.
- Reduce C_u to the sub-femtofarad level with the delta-length layout to raise the sample rate and lower power, while managing the resulting loss of range.
- Reduce comparator offset without increasing input capacitance, for example with an auxiliary trim pair.
- Develop a corner-robust cascode bias to make the kickback-reduction cascode usable.
- Extend the noise analysis to the full system, including sampling (kT/C) noise.

IX. Conclusion

A 10-bit, 5 MS/s asynchronous SAR ADC was designed and characterized at the schematic level in TSMC 65 nm CMOS with a 1 V supply. The design combines V_{CM}-based switching, bootstrapped sampling, and a self-timed two-stage dynamic comparator. The comparator completes a full conversion in 44.3 ns at the slowest corner against a 150 ns budget, with negligible hysteresis and input-referred noise below the quantization noise at the typical corner. The ADC achieves

54.77 dB SNDR, 8.81 effective bits, and 62.99 dBc SFDR, comparable to the reference design [1], at 126.4 μW. The work highlighted the central trade-offs in SAR comparator design between speed, mismatch, input capacitance, and kickback.

ACKNOWLEDGMENT

The authors thank Dr. Apisak Worapishet and Dr. Phanumas Khumsat, along with the IC design team at Analog Devices, for their guidance and mentorship throughout this project.

REFERENCES

[1] K. Pelzers, M. van der Struijk, and P. Harpe, "A 0.0022 mm^2 10 bit 20 MS/s SAR ADC with passive single-ended-to-differential-converter," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 70, no. 1, pp. 29–39, Jan. 2023, doi: 10.1109/TCSI.2022.3203792.

[2] P. J. A. Harpe, W. Dehaene, and M. Steyaert, "A 26 μW 8 bit 10 MS/s asynchronous SAR ADC for low energy radios," *IEEE J. Solid-State Circuits*, vol. 46, no. 7, pp. 1585–1595, Jul. 2011, doi: 10.1109/JSSC.2011.2143870.

[3] T. Rabuske and J. Fernandes, *Charge-Sharing SAR ADCs for Low-Voltage Low-Power Applications*. Cham, Switzerland: Springer, 2017.

[4] C.-C. Liu, S.-J. Chang, G.-Y. Huang, and Y.-Z. Lin, "A 10-bit 50-MS/s SAR ADC with a monotonic capacitor switching procedure," *IEEE J. Solid-State Circuits*, vol. 45, no. 4, pp. 731–740, Apr. 2010, doi: 10.1109/JSSC.2010.2042254.

[5] A. M. Abo and P. R. Gray, "A 1.5-V, 10-bit, 14.3-MS/s CMOS pipeline analog-to-digital converter," *IEEE J. Solid-State Circuits*, vol. 34, no. 5, pp. 599–606, May 1999, doi: 10.1109/4.760369.

[6] B. Razavi, "The bootstrapped switch [A circuit for all seasons]," *IEEE Solid-State Circuits Mag.*, vol. 7, no. 3, pp. 12–15, 2015, doi: 10.1109/MSSC.2015.2449714.

[7] P. Harpe, "A compact 10-b SAR ADC with unit-length capacitors and a passive FIR filter," *IEEE J. Solid-State Circuits*, vol. 54, no. 3, pp. 636–645, Mar. 2019, doi: 10.1109/JSSC.2018.2878830.

[8] P. Harpe, E. Cantatore, and A. van Roermund, "A 10b/12b 40 kS/s SAR ADC with data-driven noise reduction achieving up to 10.1b ENOB at 2.2 fJ/conversion-step," *IEEE J. Solid-State Circuits*, vol. 48, no. 12, pp. 3011–3018, Dec. 2013, doi: 10.1109/JSSC.2013.2278471.

[9] Analog Devices, "Histogram testing determines DNL and INL errors," Technical Article, Jun. 2003. [Online]. Available: https://www.analog.com/en/resources/technical-articles/histogram-testing-determines-dnl-and-inl-errors.html

[10] M. J. M. Pelgrom, A. C. J. Duinmaijer, and A. P. G. Welbers, "Matching properties of MOS transistors," *IEEE J. Solid-State Circuits*, vol. 24, no. 5, pp. 1433–1439, Oct. 1989, doi: 10.1109/JSSC.1989.572629.

[11] *IEEE Standard for Terminology and Test Methods for Analog-to-Digital Converters*, IEEE Std 1241-2010, Jan. 2011, doi: 10.1109/IEEESTD.2011.5692956.

[12] M. van Elzakker, E. van Tuijl, P. Geraedts, D. Schinkel, E. A. M. Klumperink, and B. Nauta, "A 10-bit charge-redistribution ADC consuming 1.9 μW at 1 MS/s," *IEEE J. Solid-State Circuits*, vol. 45, no. 5, pp. 1007–1015, May 2010, doi: 10.1109/JSSC.2010.2043893.